Alexander Dagher

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Profile: Quantitative Economics/Computer Science graduate with hands-on data engineering training (SQL, PySpark, AWS) and applied research experience in quantitative and systematic investment modeling, including forecasting, portfolio optimization, and decision-ready reporting — comfortable owning a problem end-to-end, from raw data to a client-ready analysis.

EDUCATION

St. Olaf College: Bachelor's Degree **Graduation Date:** MAY 2026

Double Major: Quantitative Economics/Computer-Science **Cumulative GPA:** 3.4/4

Skills: Python, R, Stata, SQL, Java (Spring Boot), C++, TypeScript, JavaScript, Vue3, HTML, YAML, JSON, GitLab, ADO, Google Suite, Office 365, Excel, Numpy, Pandas, Scikit-learn, XGboost, Tensorflow, Keras, Matplotlib, Seaborn, Statsmodels, Scipy, Hugging Face, REST APIs, Pinecone, Ubuntu, Linux, Qiskit, Jupiter Notebook, Prompting, Agentic AI, AI workflow optimization, CI/CD, PySpark, Apache Spark, AWS (S3, EMR, IAM, Glue), Apache Airflow, Apache Kafka, Apache Iceberg, Snowflake, Docker, ETL/Data Pipelines, Data Modeling

INTERNSHIP EXPERIENCE

Data Engineer Associate — Revature, in partnership with Cognizant | MAY 2026 - AUG 2026

- Completed an intensive data engineering apprenticeship covering Python, SQL, PySpark, and AWS/cloud data concepts, certified through Revature's technical training program.
- Served as Student Infrastructure Lead on a hospital patient data pipeline: built an AWS environment (S3, EMR, IAM), a PySpark transformation layer writing Apache Iceberg tables to a Glue Catalog, and an Airflow-orchestrated ETL DAG.
- Built a Python/SQL command-line application with full CRUD functionality and a Pandas/Matplotlib analytics layer for data visualization.

BackEnd Developer Intern — Euler Data Solutions, Bordeaux, France | JUN 2025 - AUG 2025

- Developed a Spring Boot AI microservice (Java 21) integrating a locally hosted Mistral 7B LLM for structured, safe information extraction from PDF reports.
- Built REST APIs and dynamic prompt templates with guardrails and JSON validation for reliable AI-agent behavior, and refactored project architecture using Feign clients, DTOs, and clean code principles.
- Collaborated in weekly code reviews, applied feedback for immutability, modularization, and streamlining service logic.

Business Analyst Intern — Cafezal, Milan, Italy | FEB 2025 - MAY 2025

- Designed and maintained analytics pipelines combining POS, CRM, and web data to support marketing performance tracking and segmentation analysis.
- Developed forecasting and scenario analysis tools to support sales planning, product decisions, and AOP-style planning discussions.
- Built management-ready summaries and visual dashboards to communicate KPIs, trends, and performance gaps.
- Partnered with operations and marketing teams to translate data findings into process improvements, business cases, and strategic decisions.

RESEARCH EXPERIENCE

Economics Research Fellow — St. Olaf College, Faculty: Kulsoom Hisam | SEP 2025–MAY 2026

- Automated data collection and clause coding workflows using Python and Stata for historical treaty analysis.
- Supported data quality validation and process reproducibility through consistent documentation and version control.
- Collaborated with the research team to align analytical workflows with project milestones and reporting deadlines.

Student Community Research & Data Analyst — St. Olaf College, IFC | SEP 2025–MAY 2026

- Automated large-scale data collection and analysis workflows using Python and Stata to support quantitative economic research.
- Conducted regression-based analysis and data validation to ensure statistical accuracy and reproducibility.
- Presented findings to faculty stakeholders through structured reports and visual summaries.

Hybrid Quantum-Classical Market Forecasting & Portfolio Optimization Research (Personal)

- Under guidance of **Professor Michael Haydock (IBM Fellow Emeritus)**, designed a hybrid financial modeling framework combining classical machine learning (LSTM, XGBoost) with quantum computing techniques for market forecasting and portfolio optimization. Applied predictive models to equity market data and explored quantum angle encoding / Monte Carlo methods to improve allocation decisions under uncertainty. Conducted research on integrating emerging computational methods with real-world investment strategy design.
https://github.com/laxoune/Quantum-trading

Polymarket / Kalshi Arbitrage Bot (Personal)

- Built a systematic arbitrage strategy comparing pricing across Polymarket and Kalshi prediction markets, calculating implied probabilities and spreads to identify inefficiencies after fees and execution risk. Applied API-driven data collection, probability modeling, and rules-based decision logic to a live market-making use case.
https://github.com/laxoune/arb